



Lesson 3.2 — End Behavior and Degree

Big idea: A polynomial's end behavior — what it does as $x \rightarrow \infty$ and $x \rightarrow -\infty$ — is decided by just two things: the parity of the degree and the sign of the leading coefficient.

Quick review

Even degree, positive leading: both ends go up. (Like x^2 .)

Even degree, negative leading: both ends go down. (Like $-x^2$.)

Odd degree, positive leading: down on left, up on right. (Like x^3 .)

Odd degree, negative leading: up on left, down on right. (Like $-x^3$.)

Why this works: for very large $|x|$, the leading term dominates everything else in the polynomial.

Worked example

Worked example. Describe end behavior of $P(x) = -3x^5 + 4x^2 - 1$.

Degree 5 (odd), leading coefficient -3 (negative). End behavior: as $x \rightarrow -\infty$, $P \rightarrow +\infty$; as $x \rightarrow +\infty$, $P \rightarrow -\infty$.

Warm-ups

1. State the end behavior of $P(x) = x^4 - 3x^2 + 1$.
2. State the end behavior of $P(x) = -2x^3 + 5$.
3. State the end behavior of $P(x) = 7x^6 - x$.

Practice

1. Match each polynomial to its end behavior: (a) up/up (b) down/down (c) down/up (d) up/down.
 $P_1 = 2x^3$, $P_2 = -x^4$, $P_3 = x^4$, $P_4 = -x^3$.
2. A polynomial has end behavior: as $x \rightarrow -\infty$, $P \rightarrow +\infty$ and as $x \rightarrow +\infty$, $P \rightarrow +\infty$. What can you say about its degree and leading coefficient?
3. Sketch (in words) the end behavior of $P(x) = (x - 2)(x + 1)(x - 3)$.

Challenge

1. A polynomial has odd degree and positive leading coefficient. Explain why it must have at least one real zero. (Use end behavior.)
2. Construct a polynomial with end behavior down/down and y-intercept -5 .



Answer key — Lesson 3.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Degree 4 (even), leading + $\rightarrow$ both ends UP.
2. Degree 3 (odd), leading - $\rightarrow$ up on left, down on right.
3. Degree 6 (even), leading + $\rightarrow$ both ends UP.

Practice

1. $P_1 \rightarrow c$ (down/up). $P_2 \rightarrow b$ (down/down). $P_3 \rightarrow a$ (up/up). $P_4 \rightarrow d$ (up/down).
2. Even degree, positive leading coefficient.
3. Expand mentally: leading term is x^3 (degree 3, +). End behavior: down on left, up on right.

Challenge

1. Down/up means the polynomial goes from $-\infty$ to $+\infty$ continuously; by the Intermediate Value Theorem it must cross 0 somewhere.
2. Even degree, negative leading. Example: $P(x) = -x^2 - 5$ (or $-x^4 + x - 5$).